

Time and space complexity - Big O

What is Big O ?

- Big O is a way to categorize your algorithms time or memory requirements based on input.
- It is not exact measure like how many CPU units it cost to run the algorithm.
- It is meant to be generalize the growth of your algorithm, for both computation and memory.

Why use big O ?

- It helps us determine what data structures and algorithms to use → create better program

Simplest trick for complexity

- Just look for loops over the input

3 important concepts

1. Growth is with respect to the input
2. Constants are dropped
 - Why ? Because they don't significantly impact growth rate
 - Fact: Because we don't consider constant, some $O(n^2)$ would run faster than $O(n)$ for smaller input
3. Worst case is usually the way we measure

The common complexities

Big-O Complexity

